SCH3U Final Exam - Units 1 to 4 - Teacher Version / Answer Key (verB)

Subject: chemistry
Course Code: SCH3U
Unit: Final Exam Review - Units 1 to 4

Answer Key

1. [atomic_structure | multiple_choice | 1 mark] B. Magnesium has a greater effective nuclear charge on its valence electrons.
2. [atomic_structure | multiple_choice | 1 mark] D. F-, Na+, and Mg2+ are all isoelectronic with neon.
3. [chemical_reactions | multiple_choice | 1 mark] C. (NH4)3PO4
4. [chemical_reactions | multiple_choice | 1 mark] C. CuCl2 is copper(II) chloride
5. [chemical_reactions | multiple_choice | 1 mark] A. Na2CO3(aq) + CaCl2(aq) produces insoluble CaCO3(s).
6. [chemical_reactions | multiple_choice | 1 mark] B. 2KClO3 -> 2KCl + 3O2
7. [chemical_reactions | multiple_choice | 1 mark] A. Na+ and NO3-
8. [stoichiometry | multiple_choice | 1 mark] B. 0.500 mol
9. [stoichiometry | multiple_choice | 1 mark] C. 2.67 mol
10. [stoichiometry | multiple_choice | 1 mark] B. Cl2 is limiting.
11. [stoichiometry | multiple_choice | 1 mark] C. 90.0%
12. [solutions_and_concentration | multiple_choice | 1 mark] C. 2.50 mol/L
13. [solutions_and_concentration | multiple_choice | 1 mark] B. 1.0 mol/L
14. [solubility_and_solvents | multiple_choice | 1 mark] C. Oil is non-polar while water is polar.
15. [solutions_and_concentration | multiple_choice | 1 mark] A. 0.00150 mol
16. [atomic_structure | multiple_choice | 1 mark] B. They have different numbers of neutrons.
17. [atomic_structure | multiple_choice | 1 mark] B. Mg
18. [chemical_reactions | multiple_choice | 1 mark] C. neutralization
19. [chemical_reactions | multiple_choice | 1 mark] C. K2SO4
20. [solutions_and_concentration | multiple_choice | 1 mark] B. It can still dissolve more solute.
21. [atomic_structure | written_response | 3 marks] 1s2 2s2 2p6 3s2 3p4; sulfur gains 2 electrons to achieve a stable octet like argon.
22. [chemical_reactions | written_response | 4 marks] a) BaCl2(aq) + Na2SO4(aq) -> BaSO4(s) + 2NaCl(aq); b) 2Al(s) + 3CuCl2(aq) -> 2AlCl3(aq) + 3Cu(s)
23. [chemical_reactions | written_response | 4 marks] Molecular: Ca(NO3)2(aq) + Na2CO3(aq) -> CaCO3(s) + 2NaNO3(aq); net ionic: Ca2+(aq) + CO3^2-(aq) -> CaCO3(s)
24. [chemical_reactions | written_response | 3 marks] MgCO3(s) + 2HCl(aq) -> MgCl2(aq) + H2O(l) + CO2(g); observations can include bubbling and the solid dissolving.
25. [atomic_structure | written_response | 4 marks] Aluminum's first removed electron is in the

higher-energy 3p sublevel, while magnesium loses a 3s electron. The 3p electron is easier to remove despite the greater nuclear charge.
26. [stoichiometry | written_response | 5 marks] 15.0 g Fe2O3 is about 0.0939 mol; this yields about 0.188 mol Fe, or 10.5 g Fe.
27. [stoichiometry | written_response | 5 marks] N2 is 0.500 mol and H2 is about 1.49 mol, so H2 is limiting. This yields about 0.993 mol NH3, or 16.9 g NH3.
28. [stoichiometry | written_response | 5 marks] CuSO4 + 2NaOH -> Cu(OH)2 + Na2SO4; theoretical yield is about 2.08 g Cu(OH)2, so percent yield is about 93.8%.
29. [solutions_and_concentration | application | 5 marks] 2H3PO4 + 3Ca(OH)2 -> Ca3(PO4)2 + 6H2O; n(H3PO4) = 0.00875 mol, so n(Ca(OH)2) = 0.0131 mol and c is about 0.625 mol/L.
30. [solutions_and_concentration | application | 6 marks] Need 0.0750 mol CuSO4. If using anhydrous CuSO4, mass is about 12.0 g. A strong procedure includes weighing, dissolving in a beaker, quantitative transfer to a 500.0 mL volumetric flask, rinsing the beaker and funnel, filling to the mark, and inverting to mix. Reducing concentration error can include careful meniscus reading at eye level.
31. [solubility_and_solvents | application | 6 marks] Damp filter paper makes the measured product mass too high, so percent yield appears too high. Leaving precipitate in the beaker lowers the collected mass, so percent yield appears too low. Water is suitable because it dissolves the ionic reactants so the ions can mix and form the insoluble precipitate.

Student Paper Reference

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Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Part A: Multiple Choice

Section Plan: 20 questions | 20 marks

Instructions: Choose the best answer for each question. No work is required unless you want to use the space for rough calculations.

1. Which statement best explains why magnesium has a higher first ionization energy than sodium?
A. Magnesium has fewer occupied energy levels.
B. Magnesium has a greater effective nuclear charge on its valence electrons.
C. Sodium has more protons than magnesium.
D. Sodium is more stable because it has a full valence shell. (1 mark)

2. Which species has the same electron arrangement as neon?
A. F-
B. Na+
C. Mg2+
D. All of the above (1 mark)

3. Which formula is correct for ammonium phosphate?
A. (NH4)PO4
B. NH4PO4
C. (NH4)3PO4
D. NH43PO4 (1 mark)

4. Which compound name is correct?
A. CuCl is copper(II) chloride
B. CuCl2 is copper(I) chloride
C. CuCl2 is copper(II) chloride
D. CuCl is cupric chloride (1 mark)

5. Which reaction will definitely produce a precipitate?
A. Na2CO3(aq) + CaCl2(aq)
B. KNO3(aq) + NaCl(aq)
C. HCl(aq) + NaOH(aq)
D. Mg(s) + O2(g) (1 mark)

6. Which balanced equation correctly represents the decomposition of potassium chlorate?
A. KClO3 -> KCl + O3
B. 2KClO3 -> 2KCl + 3O2
C. KClO3 -> K + Cl + 3O
D. 2KClO3 -> 2KClO + O2 (1 mark)

7. In the net ionic equation Ag+(aq) + Cl-(aq) -> AgCl(s), the spectator ions could be
A. Na+ and NO3-
B. Ag+ and Na+
C. Cl- and NO3-
D. Ag+ and Cl- (1 mark)

8. What is the amount in moles of 3.01 x 10^23 molecules of CO2?
A. 0.250 mol
B. 0.500 mol
C. 1.00 mol
D. 2.00 mol (1 mark)

9. If 4.00 mol of H2 reacts completely with excess N2 according to N2 + 3H2 -> 2NH3, how many

moles of NH_3 can form?
A. 1.33 mol
B. 2.00 mol
C. 2.67 mol
D. 4.00 mol (1 mark)

10. Which reactant is limiting when 2.0 mol Al reacts with 2.0 mol Cl_2 according to $2Al + 3Cl_2 \rightarrow 2AlCl_3$?
A. Al
B. Cl_2
C. $AlCl_3$
D. Neither reactant (1 mark)

11. A reaction has a theoretical yield of 32.0 g and an actual yield of 28.8 g. What is the percent yield?
A. 80.0%
B. 85.0%
C. 90.0%
D. 111% (1 mark)

12. What is the molar concentration of a solution made by dissolving 0.750 mol of solute to make 300.0 mL of solution?
A. 0.250 mol/L
B. 1.50 mol/L
C. 2.50 mol/L
D. 4.00 mol/L (1 mark)

13. A 25.0 mL sample of 6.0 mol/L HNO_3 is diluted to 150.0 mL. What is the new concentration?
A. 0.50 mol/L
B. 1.0 mol/L
C. 1.5 mol/L
D. 36 mol/L (1 mark)

14. Why does oil not dissolve well in water?
A. Oil molecules are ionic.
B. Water molecules are non-polar.
C. Oil is non-polar while water is polar.
D. Oil has a lower density than water. (1 mark)

15. If 15.0 mL of 0.100 mol/L HCl is neutralized by NaOH, how many moles of HCl reacted?
A. 0.00150 mol
B. 0.00667 mol
C. 0.0150 mol

D. 0.150 mol (1 mark)

16. Which statement about isotopes of the same element is correct?
A. They have different numbers of protons.
B. They have different numbers of neutrons.
C. They have different atomic numbers.
D. They have different chemical symbols. (1 mark)

17. Which Period 3 element forms a 2+ ion most readily?
A. Na
B. Mg
C. Al
D. Si (1 mark)

18. The reaction H2SO4(aq) + 2KOH(aq) -> K2SO4(aq) + 2H2O(l) is best classified as
A. synthesis.
B. decomposition.
C. neutralization.
D. single displacement. (1 mark)

19. Which compound is soluble in water according to common solubility rules?
A. AgCl
B. PbI2
C. K2SO4
D. CaCO3 (1 mark)

20. Which sample represents an unsaturated solution?
A. A solution holding the maximum dissolved solute at that temperature
B. A solution that still dissolves more solute at that temperature
C. A solution containing an insoluble precipitate only
D. A solution with equal masses of solute and solvent (1 mark)

Part B: Short Answer and Equations

Section Plan: 5 questions | 18 marks

Instructions: Show key steps. Include balanced equations, state symbols, and concise explanations where required.

21. Write the full electron configuration for sulfur, S, and explain why sulfur commonly forms a 2- ion. (3 marks)

22. Predict the products, balance the equation, and include states for each reaction.
a) BaCl2(aq) + Na2SO4(aq)

b) Al(s) + CuCl2(aq) (4 marks)

23. Write the balanced molecular equation, complete ionic equation, and net ionic equation for the reaction between aqueous calcium nitrate and aqueous sodium carbonate. (4 marks)

24. Write a balanced equation with states for the reaction of magnesium carbonate with hydrochloric acid, then state two observable signs that would indicate the reaction occurred. (3 marks)

25. Explain why aluminum has a lower first ionization energy than magnesium even though aluminum is to the right of magnesium in Period 3. (4 marks)

Part C: Problems and Calculations

Section Plan: 4 questions | 20 marks

Instructions: Show complete work, include units, and use appropriate significant digits.

26. Iron(III) oxide is reduced by carbon monoxide according to Fe2O3(s) + 3CO(g) -> 2Fe(s) + 3CO2(g). Calculate the mass of iron produced from 15.0 g of Fe2O3 if CO is in excess. (5 marks)

27. Ammonia is produced according to N2(g) + 3H2(g) -> 2NH3(g). If 14.0 g of N2 reacts with 3.00 g of H2, determine the limiting reagent and calculate the mass of NH3 produced. (5 marks)

28. Copper(II) sulfate reacts with excess sodium hydroxide to produce Cu(OH)2(s). If 3.20 g of CuSO4 is used and 1.95 g of Cu(OH)2 is collected, determine the percent yield. Write a balanced equation first. (5 marks)

29. A 35.0 mL sample of 0.250 mol/L phosphoric acid, H3PO4(aq), is completely neutralized by calcium hydroxide solution. What concentration of Ca(OH)2(aq) is required if 14.0 mL of the base is used? Write the balanced equation and show your work. (5 marks)

Part D: Long Answer and Inquiry

Section Plan: 2 questions | 12 marks

Instructions: Answer in complete sentences. Use chemistry reasoning, appropriate terminology, and clear procedure language.

30. Design a procedure to prepare 500.0 mL of 0.150 mol/L CuSO4(aq) from solid CuSO4. Your answer must include the mass calculation, major equipment, a quantitative transfer step, and one method to reduce concentration error. (6 marks)

31. A student performs a precipitation experiment by mixing Pb(NO3)2(aq) and KI(aq) to form PbI2(s), but the filter paper is still damp when the precipitate is weighed and some solid remains stuck to the beaker. Explain how each error affects the calculated percent yield, and relate your explanation to the role of water as a solvent in this experiment. (6 marks)